

Resumen del grid search

Hiperparámetros fijos:

PHASE1_EPOCHS	= 6
PHASE1_LR	= 0.001
POS_WEIGHT	= 100.0
GAUSSIAN_SIGMA	= 5.0
BATCH_SIZE	= 32
EARLY_STOPPING_PATIENCE	= 3
N_INPUT_FRAMES	= 3
INPUT_SIZE	= 640x360

Hiperparámetros barridos:

phase2_lr	= [np.float64(1e-06), np.float64(1e-05)]
phase2_epochs	= [np.int64(4), np.int64(7), np.int64(10)]

Combinaciones totales: 6

Arquitectura

Modelo	= TrackNetV2 (encoder VGG16 + decoder U-Net)
Parámetros totales (M)	= 24.9
Entrenables en fase 1 (M)	= 10.19
Congelados en fase 1 (M)	= 14.71

Resultados por combinación

tag	phase2_lr	phase2_epochs	epochs_run	best_val_loss	mean_dist_px	recall_%	fp_rate_%
lr1e-05_ep4	1e-05	4	10	0.0022	3.46	90.0	41.2
lr1e-05_ep10	1e-05	10	9	0.00198	3.88	92.4	45.4
lr1e-05_ep7	1e-05	7	9	0.00185	4.16	92.1	49.1
lr1e-06_ep7	1e-06	7	9	0.00212	5.06	90.9	55.9
lr1e-06_ep10	1e-06	10	9	0.00196	5.1	93.1	50.0
lr1e-06_ep4	1e-06	4	7	0.00222	5.34	93.4	50.9

Conclusión

Ganador: lr1e-05_ep4 (dist_media=3.46px, recall=90.0%, FP_rate=41.2%)

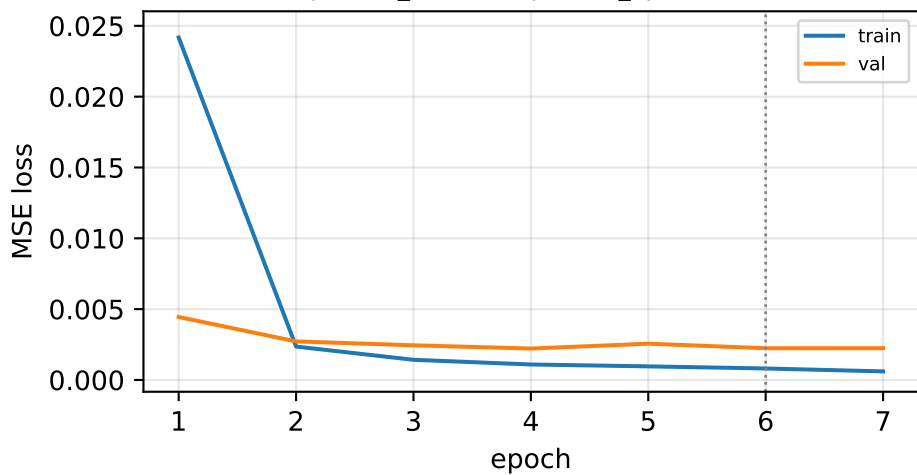
Peor combinación: lr1e-06_ep4 (dist_media=5.34px)

El ganador mejora un 35.2% la distancia media frente a la peor combinación probada.

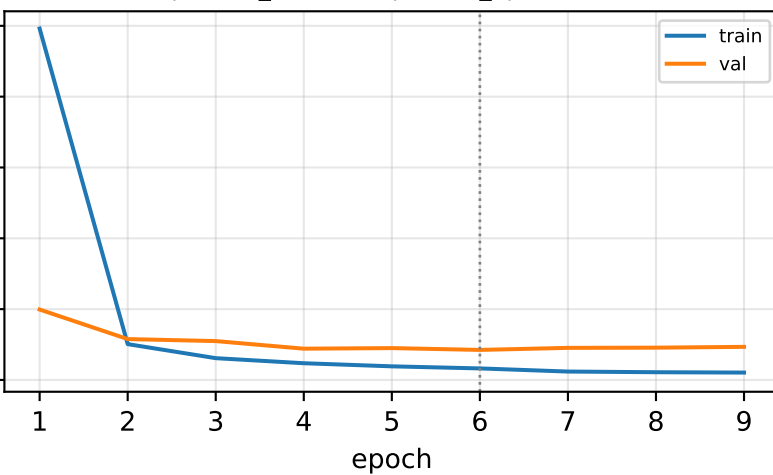
Correlación phase2_lr vs distancia media: -0.95

Correlación phase2_epochs vs distancia media: +0.05

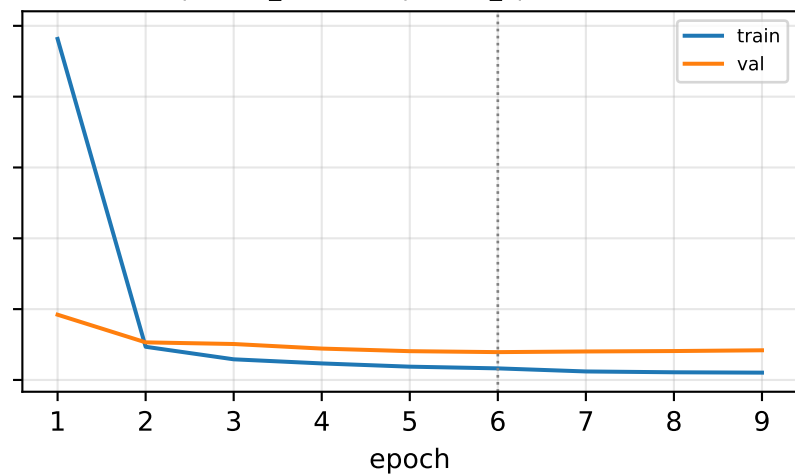
phase2_lr=1e-06, phase2_epochs=4



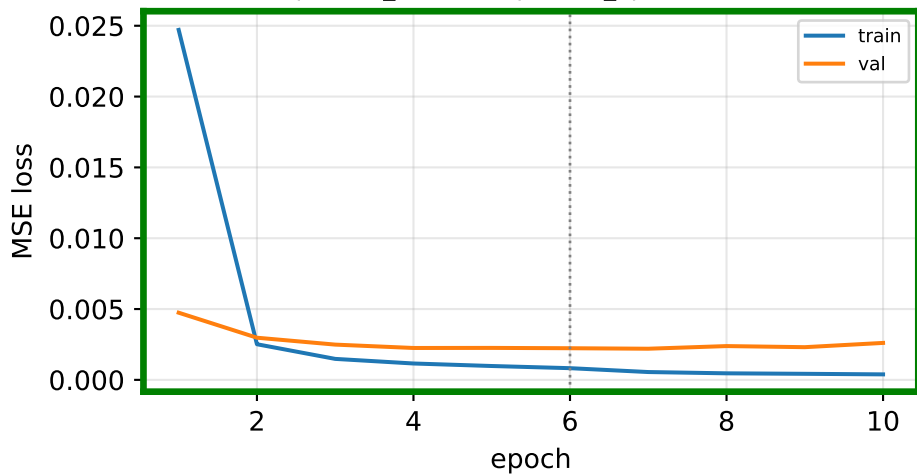
phase2_lr=1e-06, phase2_epochs=7



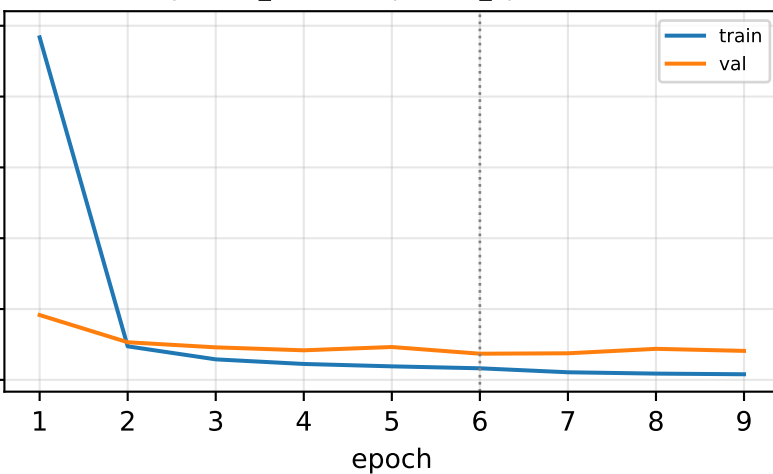
phase2_lr=1e-06, phase2_epochs=10



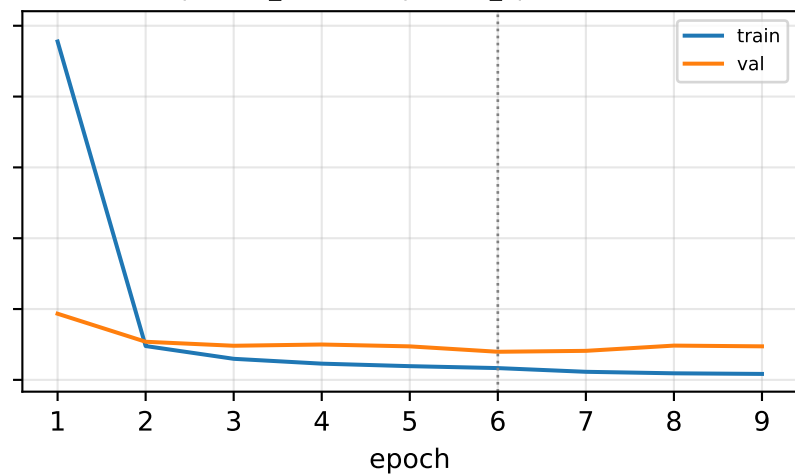
phase2_lr=1e-05, phase2_epochs=4 ★



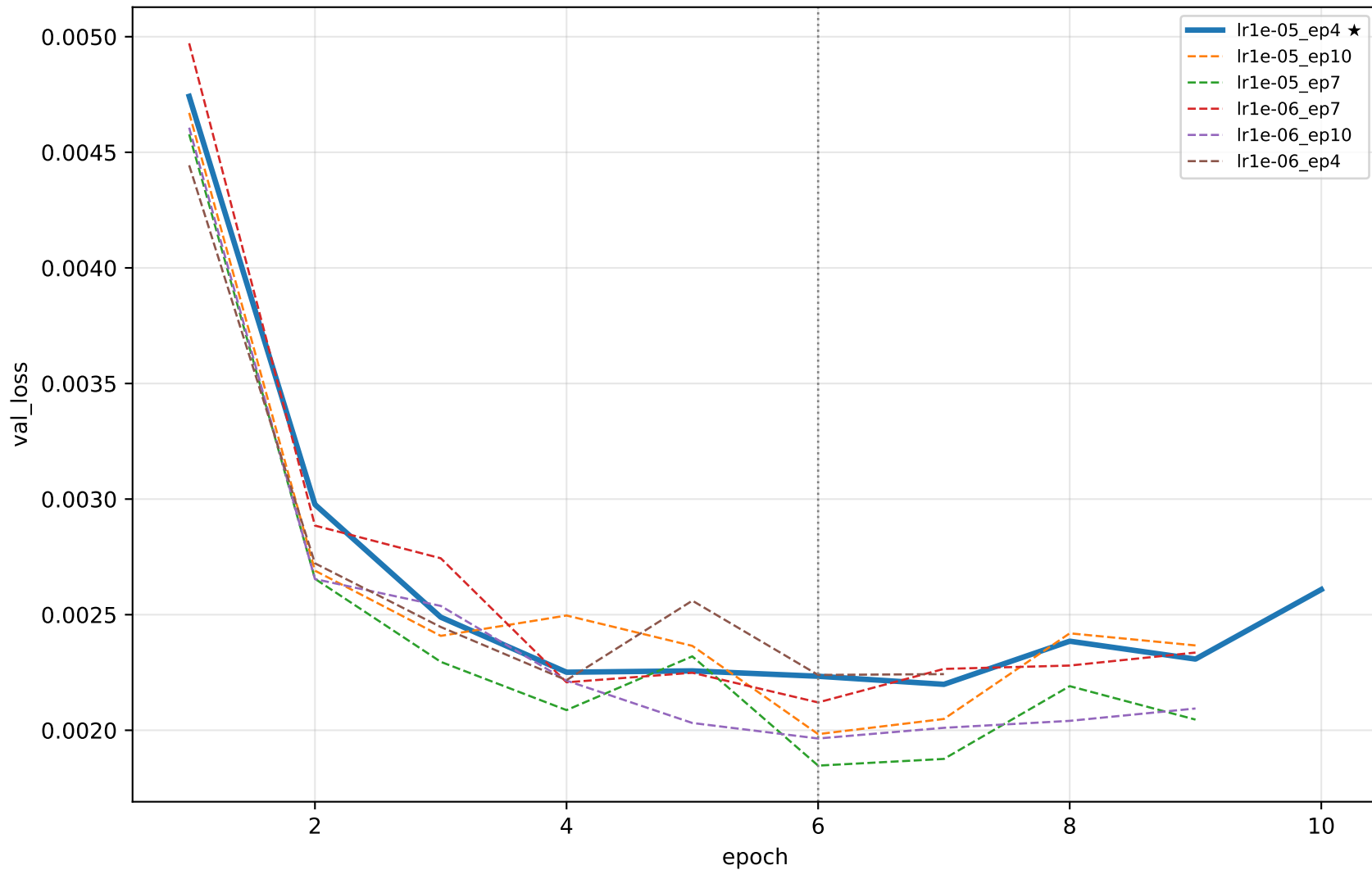
phase2_lr=1e-05, phase2_epochs=7



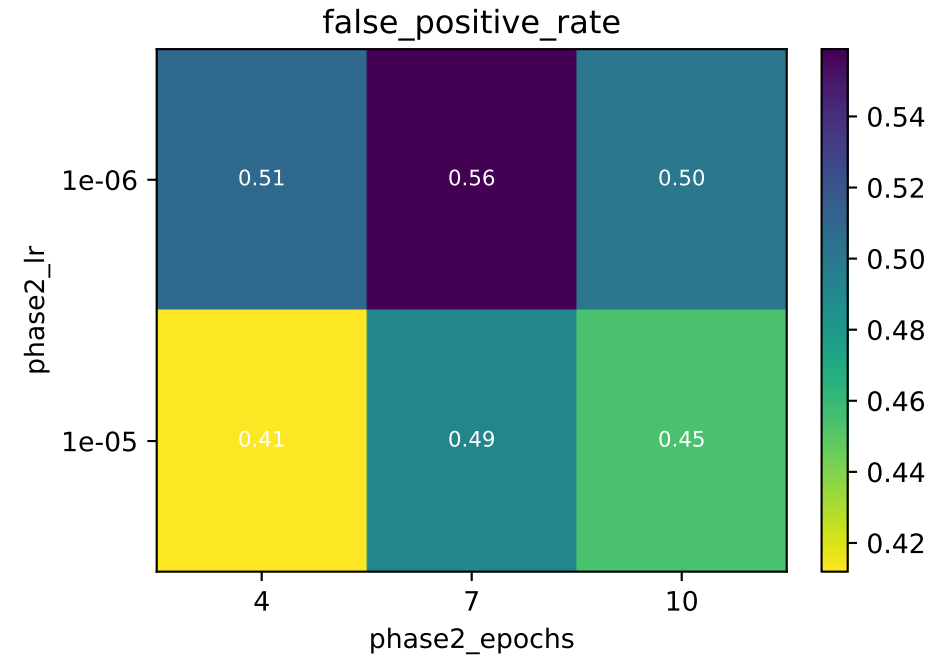
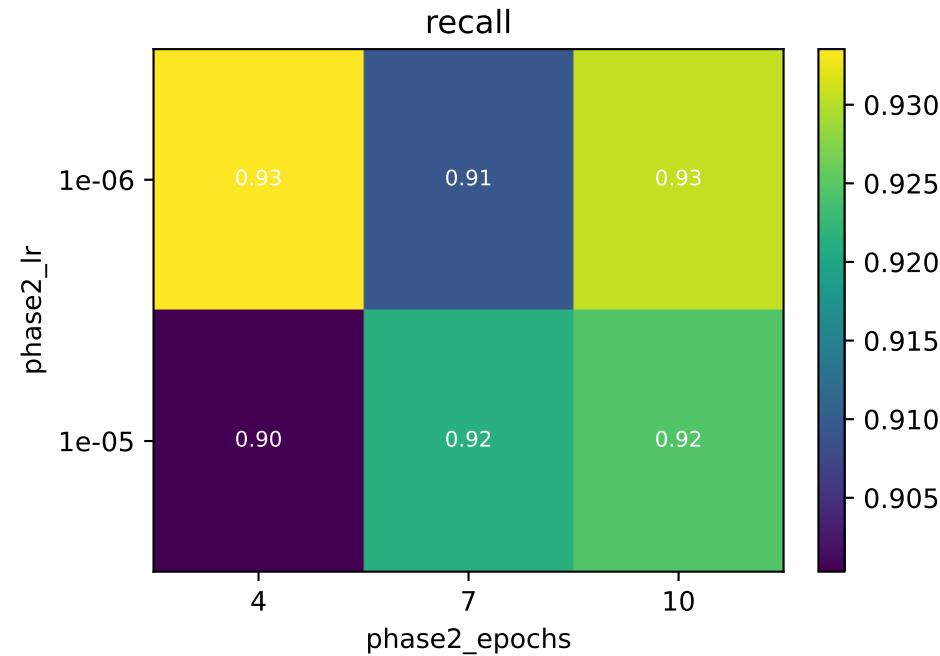
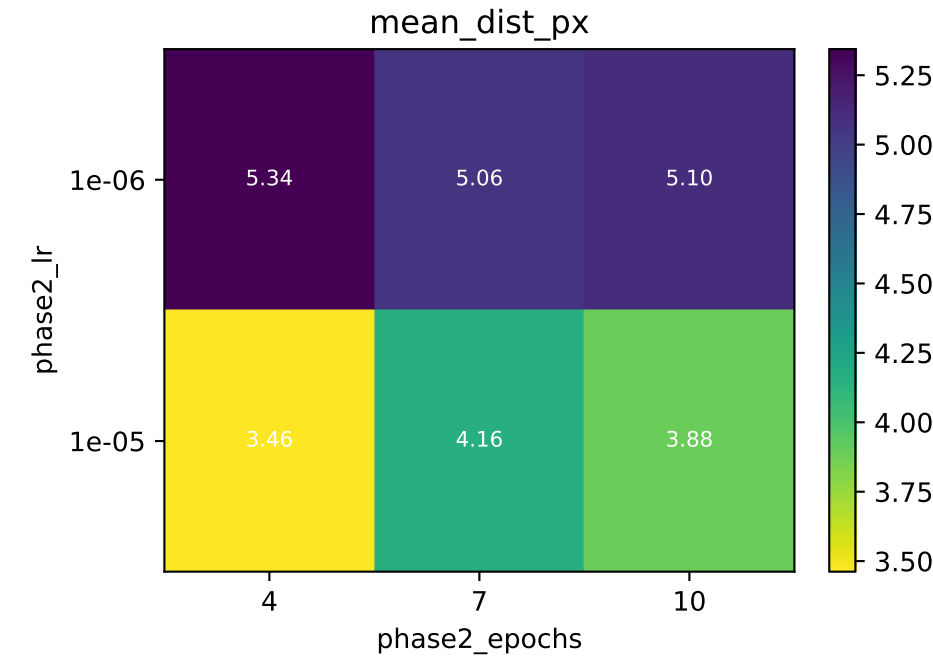
phase2_lr=1e-05, phase2_epochs=10



Val loss superpuesto — todas las combinaciones



Heatmaps de métricas sobre el espacio de hiperparámetros



Trade-off recall vs distancia (color = tasa de falsos positivos %)

